

Ergodic Wealth Dynamics, the Kelly Criterion, and the St. Petersburg Paradox

An Analytical Derivation of Log-Optimal Capital Allocation and Resolution of Infinite Expectation Lotteries

A Rigorous Analytical Derivation, Stochastic Analysis, and Empirical Verification

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Abstract

This paper presents a formal analytical derivation and stochastic evaluation of ergodic wealth dynamics and log-optimal capital allocation under the Kelly Criterion.

By contrasting ensemble-average expectations with time-average growth rates in multiplicative Bernoulli processes, we prove that maximizing long-term wealth trajectories requires optimizing the expected logarithmic growth function $g(f) = \mathbb{E}[\ln(W_t/W_0)]$.

We derive the closed-form optimal betting fraction $f^* = \frac{p(b+1)-1}{b}$, analyze parameter estimation risk justifying Fractional Kelly strategies, and clarify the historical disambiguation between Daniel Bernoulli's (1738) single-trial utility theory and John Kelly's (1956) asymptotic compounding rate.

Furthermore, we discuss applications of ergodic allocation in quantitative portfolio sizing and evaluate a companion interactive web simulation.

My Live Simulation Engine : [Access Interactive Simulation Web Application](#)

Core Math Domain: Expected Utility & Geometric Growth Dynamics

Primary Focus: Ergodic Wealth Trajectories and Log-Optimal Capital Allocation

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1 Problem Formulation & Non-Ergodic Wealth Dynamics

Consider an investor who repeatedly allocates a fixed fraction $f \in [0, 1]$ of their current wealth W_t into a sequence of independent Bernoulli investment trials.

In each trial, winning occurs with probability $p \in (0, 1)$ with odds $b > 0$, while losing occurs with probability $q = 1 - p$.

1.1 Primary Investigation Objectives

The core analytical goals of this research paper are defined across two operational domains:

- **Analytical Derivation:** Prove why multiplicative wealth growth is non-ergodic and derive the exact closed-form optimal fraction f^* that maximizes asymptotic trajectory growth.
- **Stochastic Verification:** Simulate multi-period logarithmic wealth trajectories to demonstrate how over-betting leads to absorbing ruin despite positive expected ensemble values.

1.2 Multiplicative Wealth Process Formulation

If an investor allocates fraction f , a winning outcome scales current wealth by $(1 + bf)$, whereas a losing outcome reduces capital by $(1 - f)$.

After T sequential periods with S wins and $F = T - S$ losses, cumulative wealth W_T follows a multiplicative process:

$$W_T = W_0(1 + bf)^S(1 - f)^{T-S} \quad (1)$$

Equation (1) establishes the exact state of capital at time T . Because wealth compounding is multiplicative rather than additive, expected value calculations based on ensemble averages fail to reflect individual time trajectories.

1.3 Ensemble-Average vs. Time-Average Divergence

Evaluating arithmetic expected wealth $\mathbb{E}[W_T]$ across parallel universes (ensemble average) yields an exponentially growing expectation whenever $\mathbb{E}[1 + X] > 1$:

$$\mathbb{E}[W_T] = W_0(p(1 + bf) + (1 - p)(1 - f))^T = W_0(1 + f(pb - q))^T \quad (2)$$

Equation (2) shows that expected ensemble wealth grows without bound as $T \rightarrow \infty$ whenever $pb > q$.

However, an individual operates along a single temporal path rather than across parallel universes. This non-ergodic distinction requires evaluating time-average logarithmic growth.

2 Theoretical Proof & Mathematical Derivation

To determine the true asymptotic growth rate along a single trajectory, we transform the multiplicative wealth equation into an additive process by taking the natural logarithm of wealth relative to initial capital.

$$\ln \left(\frac{W_T}{W_0} \right) = S \ln(1 + bf) + (T - S) \ln(1 - f) \quad (3)$$

Equation (3) expresses cumulative log-returns as a linear combination of winning and losing trial counts.

Dividing both sides by total periods T converts cumulative growth into an empirical time-average growth rate.

$$\frac{1}{T} \ln \left(\frac{W_T}{W_0} \right) = \left(\frac{S}{T} \right) \ln(1 + bf) + \left(1 - \frac{S}{T} \right) \ln(1 - f) \quad (4)$$

Equation (4) isolates the time-average growth rate per period.

By the Strong Law of Large Numbers, as $T \rightarrow \infty$, the sample winning proportion S/T converges almost surely to the true win probability p .

$$g(f) = \lim_{T \rightarrow \infty} \frac{1}{T} \ln \left(\frac{W_T}{W_0} \right) = p \ln(1 + bf) + (1 - p) \ln(1 - f) \quad \text{a.s.} \quad (5)$$

Equation (5) defines the expected logarithmic growth function $g(f)$. To maximize exponential wealth accumulation along a single trajectory, we must find the optimal fraction f^* that maximizes $g(f)$.

2.1 Optimization via First Principles Calculus

We differentiate $g(f)$ with respect to the allocation fraction f using the chain rule for logarithms.

$$g'(f) = \frac{d}{df} [p \ln(1 + bf) + (1 - p) \ln(1 - f)] = \frac{p \cdot b}{1 + bf} - \frac{1 - p}{1 - f} \quad (6)$$

Equation (6) evaluates the first derivative. To find the critical point f^* , we set $g'(f) = 0$.

$$\frac{pb}{1 + bf^*} = \frac{1 - p}{1 - f^*} \implies pb(1 - f^*) = (1 - p)(1 + bf^*) \quad (7)$$

Equation (7) equates both rational terms to isolate f^* .

Expanding terms across parentheses simplifies the algebraic relation.

$$pb - pbf^* = 1 + bf^* - p - pbf^* \implies pb - 1 + p = bf^* \quad (8)$$

Equation (8) cancels $-pbf^*$ from both sides and groups remaining linear terms.

Solving directly for f^* establishes the foundational Kelly Criterion.

$$f^* = \frac{p(b + 1) - 1}{b} = \frac{pb - q}{b} \quad (9)$$

Equation (9) establishes the Kelly formula: optimal fraction f^* equals expected net edge divided by win odds b .

To verify that f^* represents a global maximum, we compute the second derivative $g''(f)$.

$$g''(f) = -\frac{pb^2}{(1+bf)^2} - \frac{1-p}{(1-f)^2} < 0 \quad \forall f \in (0,1) \quad (10)$$

Equation (10) confirms that $g''(f)$ is strictly negative for all valid fractions f , proving $g(f)$ is strictly concave and f^* is a unique global maximum.

3 Computational Simulation & Asymptotic Ruin Analysis

The asymptotic growth rate function $g(f)$ exhibits asymmetric behavior around the optimal point f^* :

3.1 Capital Allocation Regimes

- **Under-allocation** ($0 < f < f^*$): Capital grows exponentially at rate $g(f) > 0$, but below the maximum theoretical rate $g(f^*)$.
- **Full Kelly Allocation** ($f = f^*$): Capital achieves the maximum possible asymptotic trajectory rate $g(f^*) = p \ln(1 + bf^*) + q \ln(1 - f^*)$.
- **Over-allocation** ($f^* < f < 2f^*$): Capital grows $g(f) > 0$, but experiences excessive volatility and lower net growth rates.
- **Critical Over-betting** ($f \geq 2f^*$): Expected growth rate becomes non-positive ($g(f) \leq 0$), driving capital to zero almost surely as $T \rightarrow \infty$.

$$\lim_{T \rightarrow \infty} W_T = 0 \quad \text{almost surely for } f \geq 2f^* \quad (11)$$

Equation (11) details the ruin boundary condition. Allocating double the Kelly fraction ($2f^*$) reduces the time-average growth rate to $g(2f^*) \leq 0$, resulting in absorbing ruin despite positive ensemble expectations.

- **Theoretical Bound:** Global maximum growth $g(f^*) = p \ln(1 + bf^*) + (1 - p) \ln(1 - f^*)$.
- **Empirical Convergence:** Numerical simulations over $T = 1,000$ steps confirm that full Kelly strategies (f^*) consistently outpace both fractional allocation ($f^*/2$) and aggressive over-allocation ($1.5f^*$).

4 Applications and Practical Observations

Log-optimal capital allocation forms a foundational pillar of modern quantitative portfolio construction, risk budgeting, and algorithmic execution:

4.1 Dynamic Position Sizing & Parameter Estimation Risk

While $f^* = \frac{pb-q}{b}$ provides the exact analytical maximum for known probability parameters, real-world deployment faces significant parameter estimation uncertainty regarding true win probability p and odds b . Overestimating p shifts allocation into the over-betting regime $f > f^*$, severely expanding drawdown volatility and exposure to ruin boundaries. Consequently, quantitative trading desks deploy Fractional Kelly strategies (such as Half-Kelly $f^*/2$), which capture 75% of theoretical growth while drastically mitigating parameter estimation risk and reducing trajectory variance by 50%.

4.2 Historical Disambiguation: Bernoulli (1738) vs. Kelly (1956)

A crucial theoretical distinction must be maintained between single-trial utility theory and multi-period asymptotic growth. Daniel Bernoulli (1738) introduced logarithmic utility $U(W) = \ln(W)$ as a static psychological resolution to the St. Petersburg Paradox—a single-period lottery with infinite expectation $\mathbb{E}[X] = \sum (1/2^k)2^k = \infty$, demonstrating that finite risk-aversion bounds subjective valuation. In contrast, John L. Kelly Jr. (1956) derived log-optimal sizing without invoking subjective risk aversion, showing that maximizing expected logarithmic growth $\mathbb{E}[\ln(W_t/W_0)]$ strictly maximizes the asymptotic compounding growth rate along a single non-ergodic temporal path in repeated multiplicative games.

4.3 Multi-Asset Portfolio Allocation

Extending single-asset Kelly to n correlated assets involves optimizing $g(\mathbf{f}) = r_f + \mathbf{f}^T \boldsymbol{\mu} - \frac{1}{2} \mathbf{f}^T \boldsymbol{\Sigma} \mathbf{f}$, where $\mathbf{f}^* = \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}$ matches mean-variance optimal portfolio weights.

5 Interactive Web Simulation Evaluation

To complement this theoretical monograph, an interactive single-file web application was developed: `kelly_wealth_sim.html`.

- **Live Interactive Application:** <https://kellycriterionandstpetersburgparado.vercel.app>
- **GitHub Source Code Repository:** <https://github.com/wheyydant/math-drafts-and-simulations>
- **Multi-Strategy Trajectory Visualizer:** Simulates and plots multi-period logarithmic wealth trajectories $\ln(W_t/W_0)$ across 1,000 temporal periods for Half-Kelly, Full Kelly, Over-Kelly, and Flat Betting strategies concurrently.
- **Real-Time Growth Curve Plot:** Dynamically renders the concave curve $g(f)$ versus fraction $f \in [0, 1]$, highlighting peak f^* and shaded ruin zones where $g(f) < 0$.
- **Academic Design Visuals:** Utilizes a high-contrast slate theme focused on data density and trajectory tracking without template visual clutter.

6 Concluding Remarks

We have presented a mathematical framework for ergodic wealth dynamics, proving that time-average capital growth in multiplicative stochastic processes requires optimizing expected logarithmic growth rather than arithmetic ensemble expectations.

The derived Kelly Criterion $f^* = \frac{pb-q}{b}$ establishes a mathematically optimal boundary for long-term growth while preventing absorption ruin.

7 Mathematical Glossary

Ergodicity:	A property of a stochastic process where time averages along a single trajectory equal ensemble averages across parallel instances. Multiplicative wealth processes are strictly non-ergodic.
Estimation Risk:	The uncertainty surrounding true distribution parameters (p, b) , requiring fractional allocation models to prevent accidental over-betting.
Time-Average Growth $(g(f))$:	The long-term rate of exponential growth per period realized along a single temporal trajectory as time $T \rightarrow \infty$.
Absorbing Ruin State:	A state $W_t \leq 0$ from which a trajectory cannot recover due to total capital depletion.